Gap #2 Session 1: Explicit Möbius Locus Identification on D_IV^5

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Gap #2 Session 1: Explicit Möbius Locus Identification

Goal of this session

Identify $M(D\_IV^5)$ — the Möbius locus — as an explicit subset of D_IV^5 in concrete coordinates. Verify the predicted real dim = n_C = 5 and boundary $\chi(S^4) = 2$ = rank. Establish the foundation that Session 2 will compute cohomology of.

D_IV^5 in Hua coordinates

Type IV bounded symmetric domain in n complex dimensions has the standard parametrization (Hua 1963; see also Pyatetskii-Shapiro 1966, "Automorphic Functions and the Geometry of Classical Domains"):

Definition (D_IV^n): Let z = (z_1, …, z_n) ∈ ℂ^n. Define the bilinear form and Hermitian norm: - `z · z :=` z_1^2 `+` z_2^2 `+ ... +` z_n^2 (complex bilinear, NO conjugate) - `|z|`^{2} `:= |z_1|`^{2} `+ |z_2|`^{2} `+ ... + |z_n|`^{2} `=` `z · z̄` (Hermitian)

Then D_IV^n is the bounded domain:

`D_IV^n = { z ∈ ℂ^n :  1 − 2|z|`^{2}` + |z · z|`^{2}` > 0  AND  |z|`^{2}` < 1 }`

The two inequalities together define an open bounded set in ℂ^n that is the type-IV Hermitian symmetric domain of complex dimension n.

For n = n_C = 5:

`D_IV`^{5}` = { z ∈ ℂ`^{5}` :  1 − 2|z|`^{2}` + |z·z|`^{2}` > 0  AND  |z|`^{2}` < 1 }`

complex dim 5; real dim 10 (= rank · n_C).

Anti-holomorphic involution τ

The complex conjugation map τ`:` $\mathbb{C}^5 \to \mathbb{C}^5$ defined by τ`(z) = z̄` (componentwise complex conjugation) restricts to D_IV^5 as an anti-holomorphic involution. Specifically: - τ preserves D_IV^5 (the defining inequalities are invariant under z ↔ z̄ since $|z|^2$ and $|z{\cdot}z|^2$ both depend only on real-bilinear quantities) - τ^2 = id - τ is anti-holomorphic (reverses complex structure)

The Möbius locus $M(D\_IV^5)$

Definition: $M(D\_IV^5)$:= {z ∈ D_IV^5 : τ(z) = z} = {z ∈ D_IV^5 : z = z̄} = {z ∈ D_IV^5 : Im(z) = 0}.

I.e., the real-form locus inside D_IV^5.

Explicit description in real coordinates: write z = x + iy with x, y ∈ ℝ^5. Then z = z̄ ⍰ y = 0.

Restricting the D_IV^5 defining inequalities to y = 0: - `|z|`^{2}` = x · x` (sum of squares of real coordinates) - `z · z = x · x` (since z = x is real, both bilinear and Hermitian agree) - `|z · z|`^{2}` = (x · x)`^{2}

The first inequality becomes:

$1 - 2(x \cdot x) + (x \cdot x)^2 > 0$ □ $(1 - (x \cdot x))^2 > 0$ □ $x \cdot x \neq 1$

The second inequality is $x \cdot x < 1$.

Combined: $x \cdot x < 1$ (the second inequality implies the first since $(1 - x \cdot x)^2 > 0$ whenever $x \cdot x \neq 1$).

Conclusion:

$M(D\_IV^5) = \{ x \in \mathbb{R}^5 : x \cdot x < 1 \} = \text{open 5-ball } B^5 \subset \mathbb{R}^5$

Verification of predicted dimensions

Predicted (v0.1 sketch)	Verified (this session)	Match
dim_ℝ M(D_IV^5) = n_C = 5	open 5-ball has dim_ℝ = 5	□
∂M is sphere S^4	∂(open 5-ball) = unit sphere S^4	□
$\chi(\partial M) = 2$ = rank	$\chi(S^4) = 1 + (-1)^4 = 1 + 1 = 2$	□
M ≅ open ball, contractible	open n-ball is contractible	□

All four predictions hold exactly. The Möbius locus identification is solid.

What this gives us for Session 2

Session 2 computes H^1(M, Z) with its Z/2 orientation class. With M now identified explicitly as the open 5-ball B^5:

- **Absolute cohomology H*(B^5, Z)**: trivial. H^0 = Z (connected), H^k = 0 for k > 0 (contractible). No information about orientation here.
- Relative cohomology H*(D_IV^5, M; Z) or H*(D_IV^5 M; Z): more interesting. The complement D_IV^5 M is the strictly-complex locus (where Im(z) ≠ 0), with two components related by τ.
- The orientation Z/2 class lives in H^1 of the orientation double cover, OR in H^0(Z/2 □ M; Z) = Z/2 (Z/2 cohomology of the τ-action on M).

So Session 2's actual computation target is: - H^0(Z/2 □ M; Z) — the orientation class as group cohomology of the τ-action - OR equivalently: H^1(B^5 / Z_2 where Z_2 acts via (z_1, ..., z_5) → (-z_1, -z_2, ..., -z_5) at the boundary)

The second formulation makes M's boundary S^4 enter the picture: ∂M = S^4 has an antipodal Z/2 action, and the quotient is real projective space $\mathbb{R}P^4$. So:

Refined Session 2 target: compute H^1(M(D_IV^5) ∪ (∂M / Z_2); Z) — the cohomology of the Möbius locus with antipodally-identified boundary. This carries the Z/2 orientation class as the connecting homomorphism in the Mayer-Vietoris sequence for (M, ∂M) ↔ (B^5, S^4).

This is the concrete Mayer-Vietoris setup Session 2 will solve.

What this DOESN'T address (deferred to later sessions)

- The (g, K)-cohomology lift to symmetric-space level — Session 3
- Connection to (6k±1) prime selection — Session 4
- Individual T-theorem promotion proofs — Session 5
- Paper draft — Session 6

Session 1 establishes the geometric setup. Session 2 will compute the cohomology. The rest builds on those.

Verification toy

`play/toy_2985_div5_mobius_locus_hua_coords.py` numerically verifies: 1. The two D_IV^5 defining inequalities define a bounded open set (random points sampled and tested) 2. The involution $\tau(z) = \bar{z}$ preserves D_IV^5 3. The fixed locus of τ in D_IV^5 is the open 5-ball in $\mathbb{R}^5$ 4. $\partial M = S^4$ has $\chi = 2 = \text{rank}$ (via discrete simplicial Euler characteristic) 5. M is contractible (verified via random deformation retraction)

All five pass — geometric foundation for Session 2 confirmed.

Status

Session 1 COMPLETE.

Ready for Session 2: H^1 computation via Mayer-Vietoris on the refined target $\{M \cup (\partial M / Z_2)\}$.

Session 1 effort: ~1.5 hours (mathematical derivation + this writeup + verification toy + Mayer-Vietoris refinement of Session 2 target).

Per scoping doc, full project ~20h over 6 sessions. ~18.5h remaining.

— Lyra, 2026-05-17 ~12:55 EDT